

$$\begin{aligned} & \frac{1}{16\pi^2} \left(-\frac{8}{81} \sum_{\mathbf{p}} g_1^4 \text{LF}_{3,0}[m_d^{\mathbf{P}}] \delta_{i1i2} \delta_{i3i4} + \frac{5}{27} \sum_{\mathbf{p}} g_1^4 \text{LF}_{4,-1}[m_d^{\mathbf{P}}] \delta_{i1i2} \delta_{i3i4} - \frac{32}{405} \sum_{\mathbf{p}} g_1^4 \text{LF}_{5,-2}[m_d^{\mathbf{P}}] \right. \\ & \quad \delta_{i1i2} \delta_{i3i4} - \frac{8}{27} \sum_{\mathbf{p}} g_1^4 \text{LF}_{3,0}[m_e^{\mathbf{P}}] \delta_{i1i2} \delta_{i3i4} + \frac{5}{9} \sum_{\mathbf{p}} g_1^4 \text{LF}_{4,-1}[m_e^{\mathbf{P}}] \delta_{i1i2} \delta_{i3i4} - \\ & \quad \frac{32}{135} \sum_{\mathbf{p}} g_1^4 \text{LF}_{5,-2}[m_e^{\mathbf{P}}] \delta_{i1i2} \delta_{i3i4} - \frac{4}{27} \sum_{\mathbf{p}} g_1^4 \text{LF}_{3,0}[m_l^{\mathbf{P}}] \delta_{i1i2} \delta_{i3i4} + \\ & \quad \frac{5}{18} \sum_{\mathbf{p}} g_1^4 \text{LF}_{4,-1}[m_l^{\mathbf{P}}] \delta_{i1i2} \delta_{i3i4} - \frac{16}{135} \sum_{\mathbf{p}} g_1^4 \text{LF}_{5,-2}[m_l^{\mathbf{P}}] \delta_{i1i2} \delta_{i3i4} - \\ & \quad \frac{4}{81} \sum_{\mathbf{p}} g_1^4 \text{LF}_{3,0}[m_q^{\mathbf{P}}] \delta_{i1i2} \delta_{i3i4} + \frac{5}{54} \sum_{\mathbf{p}} g_1^4 \text{LF}_{4,-1}[m_q^{\mathbf{P}}] \delta_{i1i2} \delta_{i3i4} - \\ & \quad \frac{16}{405} \sum_{\mathbf{p}} g_1^4 \text{LF}_{5,-2}[m_q^{\mathbf{P}}] \delta_{i1i2} \delta_{i3i4} - \frac{32}{81} \sum_{\mathbf{p}} g_1^4 \text{LF}_{3,0}[m_u^{\mathbf{P}}] \delta_{i1i2} \delta_{i3i4} + \\ & \quad \frac{20}{27} \sum_{\mathbf{p}} g_1^4 \text{LF}_{4,-1}[m_u^{\mathbf{P}}] \delta_{i1i2} \delta_{i3i4} - \frac{128}{405} \sum_{\mathbf{p}} g_1^4 \text{LF}_{5,-2}[m_u^{\mathbf{P}}] \delta_{i1i2} \delta_{i3i4} - \\ & \quad \frac{4}{27} g_1^4 \text{LF}_{3,0}[\tilde{\mu}] \delta_{i1i2} \delta_{i3i4} - \frac{2}{9} g_1^4 \text{LF}_{4,-1}[\tilde{\mu}] \delta_{i1i2} \delta_{i3i4} + \\ & \quad \frac{32}{135} g_1^4 \text{LF}_{5,-2}[\tilde{\mu}] \delta_{i1i2} \delta_{i3i4} - \frac{1}{9} g_1^4 \text{LF}_{2,1,0}[m_1, m_e^{i1}] \delta_{i1i2} \delta_{i3i4} - \\ & \quad \frac{1}{9} g_1^4 \text{LF}_{2,2,-1}[m_1, m_e^{i1}] \delta_{i1i2} \delta_{i3i4} + \frac{2}{9} g_1^4 \text{LF}_{3,1,-1}[m_1, m_e^{i1}] \delta_{i1i2} \delta_{i3i4} - \\ & \quad \frac{1}{9} g_1^4 \text{LF}_{4,1,-2}[m_1, m_e^{i1}] \delta_{i1i2} \delta_{i3i4} - \frac{1}{9} g_1^4 \text{LF}_{2,1,0}[m_1, m_e^{i2}] \delta_{i1i2} \delta_{i3i4} - \\ & \quad \frac{1}{9} g_1^4 \text{LF}_{2,2,-1}[m_1, m_e^{i2}] \delta_{i1i2} \delta_{i3i4} + \frac{2}{9} g_1^4 \text{LF}_{3,1,-1}[m_1, m_e^{i2}] \delta_{i1i2} \delta_{i3i4} - \\ & \quad \frac{1}{9} g_1^4 \text{LF}_{4,1,-2}[m_1, m_e^{i2}] \delta_{i1i2} \delta_{i3i4} - \frac{4}{81} g_1^4 \text{LF}_{2,1,0}[m_1, m_u^{i3}] \delta_{i1i2} \delta_{i3i4} - \\ & \quad \frac{4}{81} g_1^4 \text{LF}_{2,2,-1}[m_1, m_u^{i3}] \delta_{i1i2} \delta_{i3i4} + \frac{8}{81} g_1^4 \text{LF}_{3,1,-1}[m_1, m_u^{i3}] \delta_{i1i2} \delta_{i3i4} - \\ & \quad \frac{4}{81} g_1^4 \text{LF}_{4,1,-2}[m_1, m_u^{i3}] \delta_{i1i2} \delta_{i3i4} - \frac{4}{81} g_1^4 \text{LF}_{2,1,0}[m_1, m_u^{i4}] \delta_{i1i2} \delta_{i3i4} - \\ & \quad \frac{4}{81} g_1^4 \text{LF}_{2,2,-1}[m_1, m_u^{i4}] \delta_{i1i2} \delta_{i3i4} + \frac{8}{81} g_1^4 \text{LF}_{3,1,-1}[m_1, m_u^{i4}] \delta_{i1i2} \delta_{i3i4} - \\ & \quad \frac{4}{81} g_1^4 \text{LF}_{4,1,-2}[m_1, m_u^{i4}] \delta_{i1i2} \delta_{i3i4} - \frac{4}{27} g_1^2 g_3^2 \text{LF}_{2,1,0}[m_3, m_u^{i3}] \delta_{i1i2} \delta_{i3i4} - \\ & \quad \frac{4}{27} g_1^2 g_3^2 \text{LF}_{2,2,-1}[m_3, m_u^{i3}] \delta_{i1i2} \delta_{i3i4} + \frac{8}{27} g_1^2 g_3^2 \text{LF}_{3,1,-1}[m_3, m_u^{i3}] \delta_{i1i2} \delta_{i3i4} - \\ & \quad \frac{4}{27} g_1^2 g_3^2 \text{LF}_{4,1,-2}[m_3, m_u^{i3}] \delta_{i1i2} \delta_{i3i4} - \frac{4}{27} g_1^2 g_3^2 \text{LF}_{2,1,0}[m_3, m_u^{i4}] \delta_{i1i2} \delta_{i3i4} - \\ & \quad \frac{4}{27} g_1^2 g_3^2 \text{LF}_{2,2,-1}[m_3, m_u^{i4}] \delta_{i1i2} \delta_{i3i4} + \frac{8}{27} g_1^2 g_3^2 \text{LF}_{3,1,-1}[m_3, m_u^{i4}] \delta_{i1i2} \delta_{i3i4} - \\ & \quad \frac{4}{27} g_1^2 g_3^2 \text{LF}_{4,1,-2}[m_3, m_u^{i4}] \delta_{i1i2} \delta_{i3i4} + \frac{2}{9} g_1^4 \text{LF}_{2,1,0}[m_e^{i1}, m_1] \delta_{i1i2} \delta_{i3i4} - \\ & \quad \frac{1}{9} g_1^4 \text{LF}_{3,1,-1}[m_e^{i1}, m_1] \delta_{i1i2} \delta_{i3i4} + \frac{2}{9} g_1^4 \text{LF}_{2,1,0}[m_e^{i2}, m_1] \delta_{i1i2} \delta_{i3i4} - \\ & \quad \frac{1}{9} g_1^4 \text{LF}_{3,1,-1}[m_e^{i2}, m_1] \delta_{i1i2} \delta_{i3i4} + \frac{2}{9} g_1^2 \overline{y}_e^{\text{pi}1} y_e^{\text{pi}2} \text{LF}_{2,1,0}[m_l^{\mathbf{P}}, \tilde{\mu}] \delta_{i3i4} - \\ & \quad \frac{1}{9} g_1^2 \overline{y}_e^{\text{pi}1} y_e^{\text{pi}2} \text{LF}_{2,2,-1}[m_l^{\mathbf{P}}, \tilde{\mu}] \delta_{i3i4} - \frac{1}{9} g_1^2 \overline{y}_e^{\text{pi}1} y_e^{\text{pi}2} \text{LF}_{3,1,-1}[m_l^{\mathbf{P}}, \tilde{\mu}] \delta_{i3i4} + \\ & \quad \frac{1}{9} g_1^2 \overline{y}_u^{\text{pi}3} y_u^{\text{pi}4} \text{LF}_{2,1,0}[m_q^{\mathbf{P}}, \tilde{\mu}] \delta_{i1i2} - \frac{1}{18} g_1^2 \overline{y}_u^{\text{pi}3} y_u^{\text{pi}4} \text{LF}_{2,2,-1}[m_q^{\mathbf{P}}, \tilde{\mu}] \delta_{i1i2} - \\ & \quad \frac{1}{18} g_1^2 \overline{y}_u^{\text{pi}3} y_u^{\text{pi}4} \text{LF}_{3,1,-1}[m_q^{\mathbf{P}}, \tilde{\mu}] \delta_{i1i2} + \frac{8}{81} g_1^4 \text{LF}_{2,1,0}[m_u^{i3}, m_1] \delta_{i1i2} \delta_{i3i4} - \\ & \quad \frac{4}{81} g_1^4 \text{LF}_{3,1,-1}[m_u^{i3}, m_1] \delta_{i1i2} \delta_{i3i4} + \frac{8}{27} g_1^2 g_3^2 \text{LF}_{2,1,0}[m_u^{i3}, m_3] \delta_{i1i2} \delta_{i3i4} - \\ & \quad \frac{4}{27} g_1^2 g_3^2 \text{LF}_{3,1,-1}[m_u^{i3}, m_3] \delta_{i1i2} \delta_{i3i4} + \frac{8}{81} g_1^4 \text{LF}_{2,1,0}[m_u^{i4}, m_1] \delta_{i1i2} \delta_{i3i4} - \\ & \quad \frac{4}{81} g_1^4 \text{LF}_{3,1,-1}[m_u^{i4}, m_1] \delta_{i1i2} \delta_{i3i4} + \frac{8}{27} g_1^2 g_3^2 \text{LF}_{2,1,0}[m_u^{i4}, m_3] \delta_{i1i2} \delta_{i3i4} - \\ & \quad \frac{4}{27} g_1^2 g_3^2 \text{LF}_{3,1,-1}[m_u^{i4}, m_3] \delta_{i1i2} \delta_{i3i4} - \frac{1}{9} g_1^2 \overline{y}_e^{\text{pi}1} y_e^{\text{pi}2} \text{LF}_{2,1,0}[\tilde{\mu}, m_l^{\mathbf{P}}] \delta_{i3i4} + \\ & \quad \frac{5}{9} g_1^2 \overline{y}_e^{\text{pi}1} y_e^{\text{pi}2} \text{LF}_{3,1,-1}[\tilde{\mu}, m_l^{\mathbf{P}}] \delta_{i3i4} - \frac{2}{9} g_1^2 \overline{y}_e^{\text{pi}1} y_e^{\text{pi}2} \text{LF}_{4,1,-2}[\tilde{\mu}, m_l^{\mathbf{P}}] \delta_{i3i4} - \\ & \quad \frac{1}{18} g_1^2 \overline{y}_u^{\text{pi}3} y_u^{\text{pi}4} \text{LF}_{2,1,0}[\tilde{\mu}, m_q^{\mathbf{P}}] \delta_{i1i2} + \frac{11}{18} g_1^2 \overline{y}_u^{\text{pi}3} y_u^{\text{pi}4} \text{LF}_{3,1,-1}[\tilde{\mu}, m_q^{\mathbf{P}}] \delta_{i1i2} - \\ & \quad \frac{2}{9} g_1^2 \overline{y}_u^{\text{pi}3} y_u^{\text{pi}4} \text{LF}_{4,1,-2}[\tilde{\mu}, m_q^{\mathbf{P}}] \delta_{i1i2} + \frac{2}{9} g_1^4 \text{LF}_{2,1,1,-1}[m_1, m_e^{i1}, m_u^{i3}] \delta_{i1i2} \delta_{i3i4} + \\ & \quad \frac{4}{9} g_1^4 m_1^2 \text{LF}_{2,1,1,0}[m$$